

Customers' Satisfaction in a Dial-A-Ride Problem

Abstract — This paper presents a general model for a dial-a-ride problem and a Simulated Annealing approach to solve it focusing on the quality of service. The model includes several distinct cases of the real problems and an objective function that treats transportation costs and customer's inconveniences. The routes are clustered and scheduled in a separate way us-

ing specific heuristic methods. The solution method is implemented in C++ and performed over a data set based on real problems. Computational results present new best known solutions with a higher level of quality of service than a recent method found in the literature.

Keywords: Dial-a-ride; Simulated Annealing; Heuristics; Routing.

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1. Introduction

In western countries, mainly in Europe and North America, several local authorities are implementing dial-a-ride services or reviewing existent systems to attend the increasing demand. This situation can be linked to the population aging and also to a tendency for the progress of ambulatory health care services (clinics and aid to needy people) (Cordreau and Laporte, 2007).

The Dial-a-Ride Problem (DARP) shares several features with pickup and delivery problems arising in courier services. However, the people transportation is treated in DARP, and the "quality" of service becomes more important and complex. So, punctuality, traveling and waiting times and routes duration are more critical in DARP than other similar problems, as cargo transportation and courier service, for example.

DARP is known to be an NP-Hard problem (e.g. Cordeau, 2006) and several methods have been

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adopted to solve it. Cordeau (2006) and Ropke et al. (2007) point out that an exact mathematical formulation to DARP is very complex and unable to solve real size problems in a reasonable time using commercial solvers.

Heuristics and metaheuristics are frequently used to solve the DARP. Genetic Algorithms, Tabu Search, among others, easily include several conditions to accomplish the customers' satisfaction and the operational costs simultaneously (see Cordeau and Laporte, 2003; Jorgensen et al., 2007). Besides, several specific DARP heuristics also exist (see Calvo and Colorni, 2007; Coslovich et al., 2006; Toth and Vigo, 1997).

This paper presents a simple and effective alternative for solving a DARP. A general model is proposed to represent the problem, and a Simulated Annealing with other heuristics is used to treat it. This approach aims to produce operationally economical routes satisfying the customers' demand and keeping a good quality level in the offered service.

The paper is organized as follows. Section 2 presents a brief description of DARP. The proposed model is detailed in Section 3. Section 4 describes the methods used to solve the model. Computational results are presented in Section 5, and conclusions are summarized in Section 6.

2. A Brief DARP Description

Vehicle routing problems covering passengers' transport of origins to destinations are known in literature as problems of "dial-a-ride" type. The DARP consists of creating vehicles routes and schedules to transport several customers, which specify requests of pickup and delivery between specific origin and destination locations. The objective is planning a set of routes for some vehicles with minimal cost and servicing the largest possible number of customers and respecting a set of constraints (Cordeau and Laporte, 2007).

In handicapped people transportation, a pickup point matches the address where a customer should get into the vehicle, and the associated delivery point matches the address where the same customer should leave the vehicle. A *transportation request* is defined by a pickup point, a delivery point, a time window (time interval for customer attendance) for each point and the amount of conventional and wheelchairs seats necessary to cover the request.

Figure 1 shows an example of solution for the first instance (R1a) described in the Section 5. This figure considers solutions to a traditional vehicle routing problem (a) and DARP (b) approaches. Both solutions use 1 depot and 3 vehicles (routes) to attend 48 points, but the first (a) don't consider pickup and delivery points and time windows, and the second (b) attends 24 customers (transportation requests) with their pickup and delivery points and respective time windows.

Quality of service is the critical factor determining the success of Dial-a-Ride.

Figure 1(a) shows an example of classical flower shaped routes, i.e., where the routes do not intersect each other. Flower shaped routes are commonly encountered in practice for traditional vehicle routing problems.

In the Figure 1(b), we can note that in the DARP's solution the routes are fewer "organized" than in the traditional

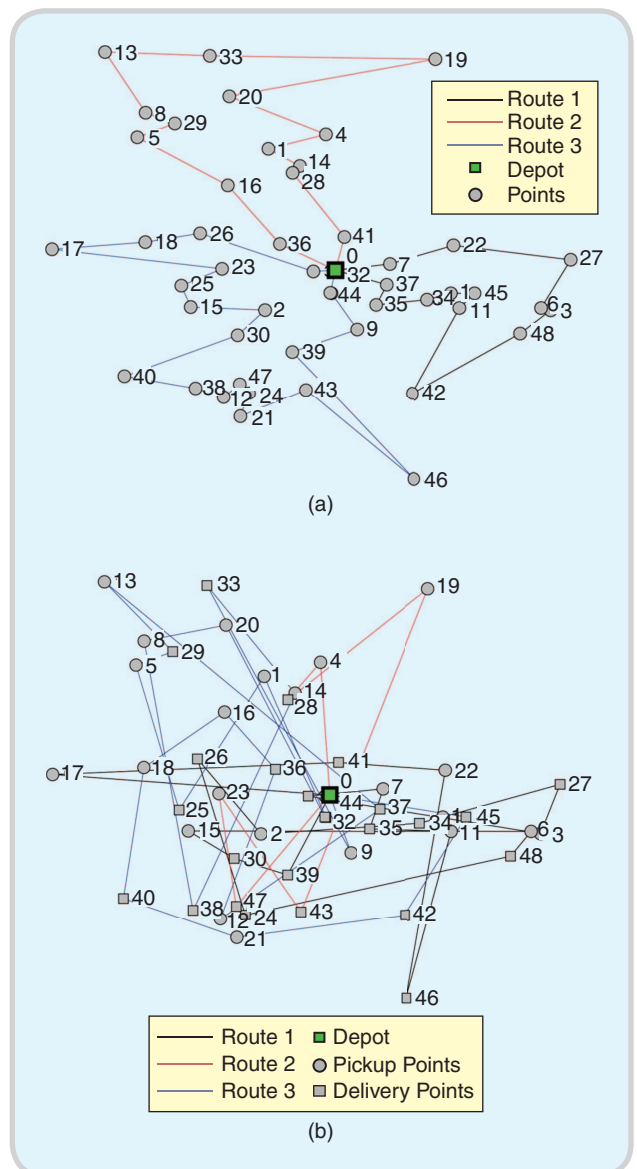


FIG 1 Solutions for a traditional vehicle routing problem and DARP.

In practice several depots with a heterogeneous set of vehicles to transport different passenger types have to be managed.

his arrival in destination $[e_{n+i}, l_{n+i}]$ (time windows). The load q_c should be a positive value in the pickup locations $i (i \in P)$ and the same value, but negative, in the respective delivery locations $n + i (n + i \in U)$.

Each vehicle $k, k \in K$, has a known capacity Q_k (amount of available seats), an associated maximum

vehicle routing, because the constraints about time windows and pickup and delivery points don't allow determining routes based only on the traveled distance. As we can see the routes are not localized, meaning that the customers are not divided into groups by points. The stronger factor seems to respect the time windows. It can also be seen that the customers very often are driven directly from their pickup to their delivery points. The level of customer service is therefore high as the customers want.

Most of the studies about DARP assumes the availability of a set of homogeneous vehicles installed in a single depot, but in practice different situations can happen, as for instance: several depots in great geographical areas and sometimes a heterogeneous set of vehicles projected to transport wheelchairs, others to transport only passengers, and others to holding both passenger types (Cordeau and Laporte, 2007).

The DARP can be considered with several different objectives: minimizing operational costs subject to satisfaction of whole demand; maximizing demand's satisfaction subject to availability of vehicles, or other combinations of these, that usually look for a balance between operational costs and quality of offered service.

3. Proposed Model

The mathematical model proposed in this paper treats the DARP with multiple vehicles, heterogeneous fleet (each vehicle has a different capacity) and multiple depots (each vehicle route begins and finishes in specific depots). This model was adapted from Jorgensen et al. (2007) and Cordeau (2006).

There exist n customers (transportation requests) to be assisted by m vehicles. Each transportation request specifies a location of pickup i and a delivery $n + i$. The following sets are defined to represent the problem:

- K : set of available vehicles ($|K| = m$).
- G^- : set of origin depots.
- G^+ : set of destination depots.
- P : set of pickup locations.
- U : set of delivery locations.
- $N = G^- \cup P \cup U \cup G^+$: set of all locations (points).

Each customer $c (c = 1, \dots, n)$ specifies the "load", q_c , necessary for his transportation (the number of seats in vehicle that will be occupied), and two time intervals, one for his departure in origin location $i [e_i, l_i]$ and another for

route duration time T_k'' , and its route begins in a specific depot g_k^- and finishes in another g_k^+ (the same or not). All of depots have their own time windows for departure (origin depot) and arrival (destination depot).

To each customer $c (c = 1, \dots, n)$ is associated a maximum ride time R_c'' (the maximum time that a customer is available to be inside the vehicle). For each location $i, i \in P \cup U$, is associated a maximum waiting time W_i'' , which is the maximum time that vehicles can be waiting until the service start. A service is the pickup or delivery of a customer in a certain location, and a time s_i is necessary to finish it. Finally, the locations of depots and points of transportation requests, and the distances d_{ij} and traveling time t_{ij} between the points i and $j, i, j \in N$ and $i \neq j$, are known in advance.

The objective is minimizing the operational costs and the customers' inconvenience (minimizing the non-essential requirements of the problem). These requirements are related to the total distance traveled by the vehicles, the number of vehicles used in problem solution, the routes duration, the customers' ride time and the waiting times in pickup and delivery locations. A set of essential requirements defines feasible solutions to the problem. These requirements are:

- The route duration performed by the vehicle k should not exceed the allowed maximum time T_k'' .
- The ride time of customer $c (c = 1, \dots, n)$ should not exceed the allowed maximum ride time R_c'' ;
- The waiting time in location $i, i \in P \cup U$, should not exceed the allowed maximum waiting time W_i'' ;
- The vehicles capacity $Q_k, k \in K$, cannot be exceeded at any location;
- The start of service in all locations $i, i \in N$, should be inside the intervals $[e_i, l_i]$.

Using these information, the decision variables can be defined as follows.

- A_i is the arrival time in location $i, i \in N$, by the vehicle that will attend it, $A_i = 0$ if $i \in G^-$ and $A_i = D_{i-1} + t_{i-1,i}$ if $i \in P \cup U \cup G^+$;
- D_i is the departure time of location $i, i \in N$, by the vehicle that will attend it, $D_i = 0$ if $i \in G^+$, $D_i = B_i + s_i$ if $i \in P \cup U$ and $D_i = B_i$ if $i \in G^-$;
- B_i represents the service start time in location $i, i \in N$, $B_i = D_i$ if $i \in G^-$ and $B_i = \max\{e_i, A_i\}$ if $i \in P \cup U \cup G^+$;

- the waiting time before service start in location $i, i \in N$, is $W_i, W_i = 0$ if $i \in G^-$ and $W_i = B_i - A_i$ if $i \in P \cup U \cup G^+$;
- Q_i indicates the load (number of busy seats) of the vehicle that attends the location $i, i \in N$, after service conclusion, $Q_i = 0$ if $i \in G^- \cup G^+$ and $Q_i = Q_{i-1} + q_i$ if $i \in P \cup U$;
- R_c is the ride time of customer c ($c = 1, \dots, n$), and $R_c = B_{n+i} - D_i$, considering i and $n+i$ be the pickup and delivery locations for the customer c .

Finally, $x_{i,j}^k = 1$ if vehicle k travels from location i to location j and $x_{i,j}^k = 0$ otherwise. The mathematical model is given as follows:

Minimize:

$$\omega_0 \sum_{k \in K} \sum_{i \in N} \sum_{j \in N; j \neq i} (d_{i,j} x_{i,j}^k) + \omega_1 \sum_{k \in K} \sum_{j \in P} x_{g_k^+, j}^k + \omega_2 \sum_{k \in K} (B_{g_k^+} - D_{g_k^-}) + \omega_3 \sum_{c=1, \dots, n} R_c + \omega_4 \sum_{i \in P \cup U} W_i + \quad (1)$$

$$\beta_0 \sum_{k \in K} \max\{0, (B_{g_k^+} - D_{g_k^-}) - T_k^n\} + \beta_1 \sum_{c=1, \dots, n} \max\{0, R_c - R_c^n\} + \beta_2 \sum_{i \in P \cup U} \max\{0, W_i - W_i^n\} + \quad (2)$$

$$\beta_3 \sum_{k \in K} \max\left\{0, \left(Q_i \sum_{i \in P \cup U} \sum_{j \in P \cup U; j \neq i; j \neq n-i} x_{i,j}^k\right) - Q_k^n\right\} + \beta_4 \sum_{i \in N} (\max\{0, e_i - B_i\} + \max\{0, B_i - l_i\}) \quad (3)$$

Subject to:

$$\sum_{j \in P \cup \{g_k^+\}} x_{g_k^+, j}^k = 1 \quad \forall k \in K \quad (4)$$

$$\sum_{i \in U \cup \{g_k^-\}} x_{i, g_k^-}^k = 1 \quad \forall k \in K \quad (5)$$

$$\sum_{k \in K} \sum_{j \in P \cup U; j \neq i} x_{i,j}^k = 1 \quad \forall i \in P \quad (6)$$

$$\sum_{j \in P \cup U; j \neq i} x_{i,j}^k - \sum_{j \in P \cup U \cup \{g_k^-\}; j \neq i; j \neq n+i} x_{n+i, j}^k = 0 \quad \forall k \in K; i \in P \quad (7)$$

$$\sum_{j \in P \cup U \cup \{g_k^-\}; j \neq i; j \neq n+i} x_{j, i}^k - \sum_{j \in P \cup U; j \neq i} x_{i, j}^k = 0 \quad \forall k \in K; i \in P \quad (8)$$

$$\sum_{j \in P \cup U; j \neq i} x_{j, i}^k - \sum_{j \in P \cup U \cup \{g_k^-\}; j \neq i; j \neq n-i} x_{i, j}^k = 0 \quad \forall k \in K; i \in U \quad (9)$$

$$B_j = (B_i + s_i + t_{i,j} + W_j) \sum_{k \in K} x_{i,j}^k \quad \forall i, j \in N; i \neq j \quad (10)$$

$$Q_j = (Q_i + q_j) \sum_{k \in K} x_{i,j}^k \quad \forall i, j \in N; i \neq j \quad (11)$$

$$A_i = B_i - W_i \quad \forall i \in P \cup U \cup G^+ \quad (12)$$

$$D_i = B_i + s_i \quad \forall i \in P \cup U \cup G^- \quad (13)$$

$$R_c = B_{n+i} - D_i \quad \forall c = 1, \dots, n; i = c \quad (14)$$

$$A_{g_k^-} = D_{g_k^+} = Q_{g_k^-} = Q_{g_k^+} = W_{g_k^-} = 0 \quad \forall k \in K \quad (15)$$

$$A_i, W_i, B_i, D_i, Q_i \geq 0 \quad \forall i \in N \quad (16)$$

$$R_c \geq 0 \quad \forall c = 1, \dots, n \quad (17)$$

$$x_{i,j}^k \in \{0, 1\} \quad \forall k \in K; \forall i, j \in N; i \neq j \quad (18)$$

The objective function is divided in two parts (expression 1 and expressions 2 and 3). The first part (1) seeks to minimize non-essential requirements. The terms of (1) represent, respectively, the total distance traveled by the vehicles, the number of vehicles used to assist all the customers, the total time of routes, the total ride time and the total waiting time.

The second part (2)–(3) seeks to minimize the violations in essential requirements. These equations include the time that exceeds the maximum time of routes duration, the total time that exceeds the maximum ride time, the total time that exceeds the maximum waiting time, the excess in vehicles capacity and the total violations on the time windows.

The non-essential requirements are penalized through a vector of positive integers (weights) $\omega = [\omega_0, \omega_1, \omega_2, \omega_3, \omega_4]$, and the essentials are penalized through a similar vector $\beta = [\beta_0, \beta_1, \beta_2, \beta_3, \beta_4]$.

Constraints (4) and (5) ensure, respectively, that each vehicle will leave its origin depot and will arrive at its destination depot (each route will be begun at origin depot and will finish at its destination depot). Constraint (6) sets that each customer will be assisted once, for just a vehicle, and constraint (7) ensures that a delivery and respective pickup locations will always be in the same route. The flow conservation is guaranteed by constraints (8) and (9). Constraint (10) sets the time at the beginning of service, the waiting time in each location and the vehicle that will assist it.

As shown in Cordeau (2006), these constraints (4)–(10) guarantee the sub-tours elimination. Constraint (11) sets the vehicles' load in each location, and constraints (12), (13) and (14) guarantee, respectively, a correct computation of arrival times, departure times, and ride times of customers. Constraint (15) "initializes" some variables about the depots. Finally, constraint (18) just guarantees that decision variables $x_{i,j}^k$ will be binary, and constraints (16) and (17) guarantee that other variables will be non-negatives.

In the proposed model, the restrictions (10) and (11) are non-linear. However, these restrictions can be linearized through equations (19), (20), (21) and (22), as presented by Cordeau and Laporte (2003) and Jorgensen et al. (2007). In these equations, M is a very large value, $M \gg B_i + s_i + t_{i,j} + W_j - B_j + Q_i + q_j - Q_j$.

$$B_j \geq B_i + s_i + t_{i,j} + W_j - M \left(1 - \sum_{k \in K} x_{i,j}^k\right) \quad (19)$$

$$B_j \leq B_i + s_i + t_{i,j} + W_j + M \left(1 - \sum_{k \in K} x_{i,j}^k\right) \quad (20)$$

$$Q_j \geq Q_i + q_j - M \sum_{k \in K} x_{i,j}^k \quad (21)$$

$$Q_j \leq Q_i + q_j + M \sum_{k \in K} x_{i,j}^k \quad (22)$$

The proposed model can be easily adapted to solve other models already known for DARP (models with single depot,

Optimization is initialized based on distribution and programming heuristics.

1. **CREATE** (m empty routes, and assign it to m vehicles);
2. **CREATE** (a list L with all customers' transportation requests);
3. **FOR** (each route k , $k = 1, 2, \dots, m$) **DO**
4. **SELECT** ($\lfloor n/m \rfloor$ requests of L);
5. **FOR** (each selected request) **DO**
6. $\text{Pos1} \leftarrow$ any position of route k ;
7. $\text{Pos2} \leftarrow$ any position of route k , but after to Pos1 ;
8. **INSERT** (the pickup point in Pos1);
9. **INSERT** (the delivery point in Pos2);
10. **END-FOR**;
11. **INSERT** (the origin depot of vehicle k in beginning of route k);
12. **INSERT** (the destination depot of vehicle k at the end of route k);
13. **REMOVE** (the requests selected of L);
14. **END-FOR**;

FIG 2 Distribution heuristic.

1. $B_0 \leftarrow e_0$; $D_0 \leftarrow B_0$;
2. **COMPUTE** (A_i, B_i, W_i, D_i, Q_i for each point $v_i \in V_k$ and $v_i \neq v_0$);
3. **COMPUTE** (F_0);
4. $B_0 \leftarrow e_0 + \min \left\{ F_0, \sum_{0 < p \leq z} w_p \right\}$; $D_0 \leftarrow B_0$;
5. **UPDATE** (A_i, B_i, W_i, D_i for each point $v_i \in V_k$ and $v_i \neq v_0$);
6. **COMPUTE** (R_i for each point $v_i \in V_k$ and $v_i \in P$);
7. **FOR** (each point $v_i \in V_k$ and $v_i \in P$) **DO**
8. **COMPUTE** (F_i);
9. $B_i \leftarrow B_i + \min \left\{ F_i, \sum_{i < p \leq z} w_p \right\}$;
10. $D_i \leftarrow B_i + s_i$; $W_i \leftarrow B_i - A_i$;
11. **UPDATE** (A_i, B_i, W_i, D_i for each point $v_j \in V_k$ and v_j after to v_i);
12. **UPDATE** (R_j for each point $v_j \in V_k$, $v_j \in P$ and v_{n+j} after to v_i);
13. **END-FOR**;

FIG 3 Programming heuristic.

just one vehicle, homogeneous fleet, and models with several different objectives). This flexibility in the choice of final objective is given by using an objective function with weights to determine the importance of each requirement in the problem solution.

As suggested in Cordeau (2006), this model makes a reduction on the number of existent arcs (paths) among

the locations (it can be seen in the mathematical formulation). Such fact is due to DARP characteristics that don't allow an arc linking a location to itself, a delivery to be accomplished before its respective pickup, and other similar characteristics.

Unlike the model proposed by Jorgensen et al. (2007), that uses variables about the locations in an exclusive mode for each vehicle ($A_i^k, B_i^k, D_i^k, W_i^k, Q_i^k, R_i^k$), unique variables are used in the proposed model ($A_i, B_i, D_i, W_i, Q_i, R_i$), reducing significantly the number of decision variables for the model. This approach is based on a compact two-index formulation proposed by Ropke et al. (2007).

4. Simulated Annealing

Simulated Annealing (SA) is a local search method that accepts worsening moves to escape local optimal. It was proposed originally by Kirkpatrick et al. (1983), and it is based on an analogy with thermodynamics, when simulating the cooling of a set of heated atoms. The SA control parameters are the cooling rate α , the iterations number for each temperature SA_{max} , the initial temperature T_0 and the freezing temperature T_c .

In this paper, the initial solution is formed through a *distribution heuristic*, which is responsible for routing the vehicles, forming clusters of locations in the routes and determining their attending sequence. This heuristic is presented in Figure 2.

In the distribution heuristic, m empty routes are created and a specific vehicle is assigned to each one. The customers' transportation requests (pickup points and its respective delivery ones) are randomly distributed on these routes, and the n transportation requests are equally divided among the m vehicles.

The insertion positions in a route are also selected at random, but always respecting the precedence constraint from the customers' transportation request (the pickup locations should always be before the delivery location). Finally, the origin and destination depots for the vehicles (associated to each route) will be

allocated as the first and last positions for all of the created routes.

The distribution heuristic treats only the vehicles' routing, and the vehicles' programming should be made to determine the arrival times in locations, the departure times, and so on. The *programming heuristic* (Figure 3) is adapted from the one presented in Cordeau and

Laporte (2003), and performs the programming trying to reduce the violations in time windows, in routes duration and ride times. A concept of “delay” is used in this heuristic. This concept was initially proposed by Savelsbergh (1992), and consists basically of delaying, as much as possible, the departure time of the origin depot and start of service in pickup locations.

Initially, the departure time from the origin depot is set as the start time of the respective time window. So, other computations (arrival time, start of service, waiting time, departure time and load of vehicle) are performed for all the next points in route. Later, the delay is computed for departure of origin depot, and the departure time is adjusted without increasing the time windows violations. An updating in the times for all the next points to origin depot is made, and the customers’ ride times are computed. The delay for each pickup point in route is computed, and the service start is adjusted, reducing the route duration and the customers’ ride time without increasing the violations in time windows. Finally, the times of all the next points and the customers’ ride time whose delivery is after the pickup point are updated.

Considering a route $V_k = \{v_0, v_i, v_j, v_{n+i}, v_{n+j}, \dots, v_z\}$ performed by the vehicle $k, k \in K, v_0$ and v_z can be defined, respectively, by the origin and destination depots of $k (v_0 \in G^-; v_z \in G^+)$. The other points represent the pickup and delivery points. The point v_{n+i} represents the delivery point corresponding to the pickup point $v_i (v_i \in P \text{ and } v_{n+i} \in U)$. For any route V_k , the delay is computed as presented in equations (23), (24) and (25).

$$F_i = \min_{i \leq j \leq z} \left\{ \sum_{i < p \leq j} W_p + (\min\{l_j - B_j, R_i'' - R_j\})^+ \right\} \text{ if } i \in P \quad (23)$$

$$F_i = \min_{i \leq j \leq z} \left\{ \sum_{i < p \leq j} W_p + (l_j - B_j) \right\} \text{ if } i \in G^- \quad (24)$$

$$F_i = 0 \text{ if } i \in U \cup G^+ \quad (25)$$

In previous equations (23)–(25), F_i represents the maximum time that vehicle (that assists the point i) could be waiting before the service beginning in point i . In equation (23), the sum of W_p indicates the total waiting time in locations assisted by the route and the following term indicates the “smaller” violation between the time windows and ride times. This equation is just defined for pickup points.

In equation (24), the violation is not considered in ride times because such equation is just defined for the origin depots, and the equation (25) is only defined for delivery points and destination depots, where delay are not considered. As presented by Savelsbergh (1992), this

Three route optimization moves are conducted in a local neighborhood: Re-order route, Re-allocate points, and Swap points.

“delay” seeks to extend the departure time of origin depot and the service beginning in the pickup points, avoiding violations in time windows, routes duration and ride times.

Three different moves were used in a neighborhood structure: *Re-order route*, *Re-allocate points* and *Swap points*. These moves are based in others often found in papers about DARP (see Cordeau and Laporte, 2003; Jorgensen et al., 2007; Savelsbergh, 1992). The depots are not considered in these moves.

The *Re-order route* move consists basically in selecting a position on a route, select a new position for this point and swap the positions. This move is shown in Figure 4. The selected point can be a pickup or a delivery point. In first case (Figure 4a), the new point must be positioned before

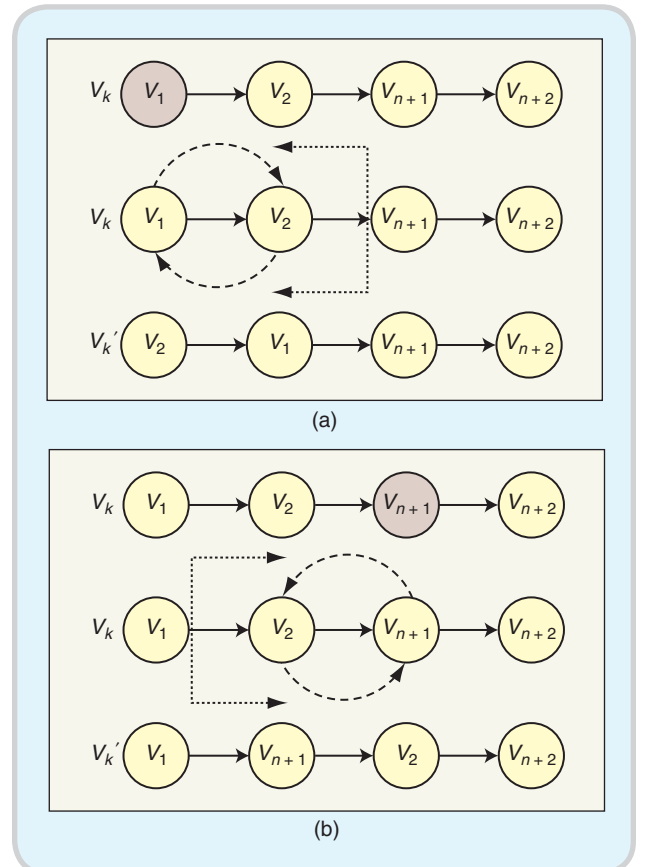


FIG 4 Re-order route.

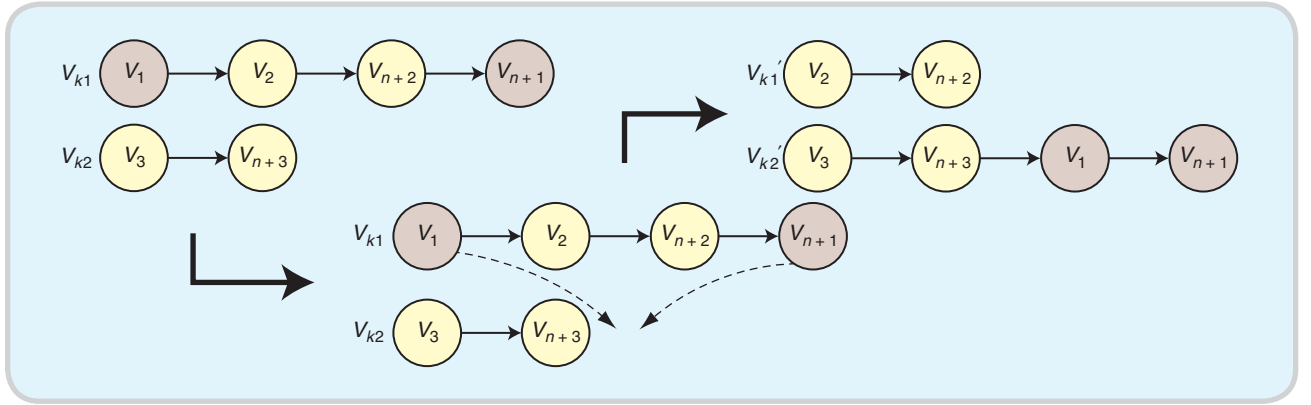


FIG 5 Re-allocate points.

its respective delivery point. In second case (Figure 4b), the new point should be placed after the respective pickup point. These “limits” are presented through the stippled lines in Figure 4.

The *Re-allocate points* move consists of selecting two routes from solution (V_{k1} and V_{k2}) and a transportation request (v_1 and v_{n+1}) in one of these routes. The selected transportation request (pickup and delivery points) is extracted from origin route (V_{k1}) and inserted in any position into the other route (V_{k2}). This move is shown in Figure 5. The pickup point (v_1) and its respective delivery point (v_{n+1}) are extracted simultaneously. The insertion of these points in “another” route is done in a separate way in any position of the route always respecting the precedence constraint.

The *Swap points* move consists of selecting any two transportation requests on any two routes, and swaps them. This move is shown in Figure 6. In this case, transportation requests are swapped, so their pickup and

delivery points are swapped simultaneously (in pairs), ensuring that pickup point will always be before its respective delivery point.

Starting from this neighborhood structure the SA was implemented using one of these movements for each neighbor solution generation. The move choice is done at random, making possible a good diversity and consequently a good solution space exploration. The objective function $f(S)$ used to evaluate the solutions is described by expressions (1), (2) and (3) (see Section 3) and the constraints presented in the model proposed in Section 3 (4 to 18) are implicitly respected in the initial heuristics and neighborhood structures described in this section.

5. Computational Results

Several experiments were performed on data instances available at *OR-Library* (<http://people.brunel.ac.uk/~mastijb/jeb/info.html>). This data set was generated by Cordeau and Laporte (2003) according to realistic assumptions provided by the Montreal Transit Commission (MTC). Twenty instances are available, and they are combined among 24 to 144 transportation requests (48 to 288 points) and 3 to 13 vehicles. The first instances (R1a–R10a) are formed by wide time windows while the remaining ones (R1b–R10b) are formed by narrow time windows. As mentioned by Cordeau and Laporte (2003), in the first group, narrow time windows were generated by first choosing a uniform random number e_i in the interval $[60, 480]$ and then choosing a uniform random number l_i in the interval $[e_i + 15; e_i + 45]$, i.e., the time windows vary from 15 to 45 minutes. In the second group, wider time windows were created by choosing the random numbers e_i and l_i in the intervals $[60, 480]$ and $[e_i + 30; e_i + 90]$, respectively, i.e., the time windows vary from 30 to 90 minutes.

The parameters used by SA in all the experiments were defined empirically: $T_0 = 20000$, $\alpha = 0.975$, $T_c = 0.01$ and $S_{max} = 1000$. The weights for the non-essential requirements (vector ω) applied in the first part of

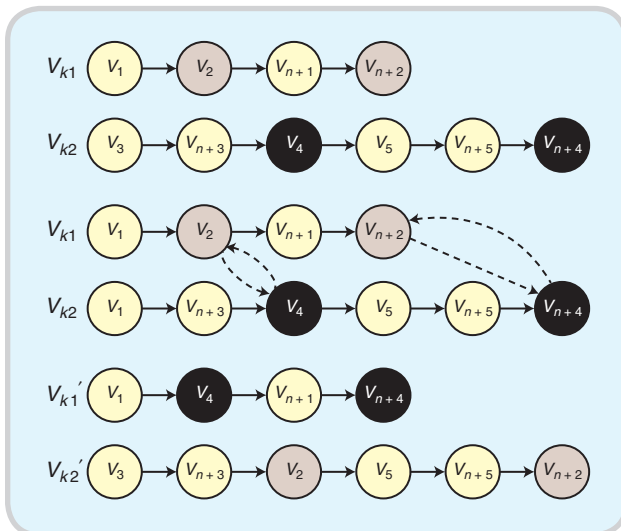


FIG 6 Swap points.

the objective function (1) were exactly the same as the ones used by Jorgensen et al. (2007). However, the weights used for violations in essential requirements (vector β) were bigger, trying to avoid getting invalid solutions for the problem. The used weights were: $\omega = [8, 0, 1, 3, 1]$ and $\beta = [1500, 1500, 1500, 1500, 1500]$.

$$\text{Deviation} = \left(\frac{f(S) \text{ average} - \text{Best } f(S)}{\text{Best } f(S)} \right) * 100 \quad (26)$$

Table 1 presents a summary of the computational results ($f(S)$ is the value of objective function). The column *Better $f(S)$* indicates the value of objective function for the best solution found in 5 tests (for each instance). The column *$f(S)$ average* presents the arithmetic average of the objective functions values found from several SA running, and the column *Deviation* presents the deviation obtained among these tests (see equation 26). In all the tests, the number of used vehicles was the same to the available, all the transportation requests were assisted, and the most important, all the obtained solutions were valid (with all the essential constraints satisfied).

The best obtained solutions (Table 2) can be directly compared with the results of Jorgensen et al. (2007) through a Genetic Algorithm (Table 3). Jorgensen et al. (2007) doesn't report results for all instances, so the comparison was made only on a subset of available instances. The results presented in Table 3 have been obtained from Jorgensen et al. (2007) original paper. The SA presents a reduction of 11.71% in the routes duration. In the average waiting time, our SA got a reduction of 48.53%. Our SA got still a reduction of 75.33% for the average customers' ride time, and the processing time was reduced in 95.37%.

All the tests were performed in a laptop with *Intel Celeron*® of 2.0 GHz processor and 256 MB of RAM memory, and the code was implemented in C++. The solutions obtained by Jorgensen et al. (2007) were performed in a PC with *Intel Celeron*® of 2.0 GHz processor and their code was implemented in Java.

The Dial-a-Ride model aims to balance customer inconvenience against operational cost according to predefined weights.

Conclusion

This paper presented a new approach to solve a dial-a-ride problem. The proposed model was able to represent the DARP in a generalized way. This approach aims at reducing the customer's inconvenience, but it can also be used to reduce operational costs. The objectives are highlighted through weights values.

The clustering and routing were performed by a distribution heuristic and neighborhood structure, and the vehicles' programming was performed by a programming heuristic. The SA with the heuristics presented in Section 4 was able to get valid solutions for the problem in all the cases with low processing time. The neighborhood structure had been shown to be appropriate and efficient for the exploration in the space of solutions.

Table 1. Summary of performed tests.

Instance	Vehicles Number	Customers Number	$f(s)$ average	Best $f(s)$	Deviation (%)
R1a	3	24	3721.57	3677.91	1.19
R2a	5	48	7101.87	7017.34	1.20
R3a	7	72	11982.18	11873.76	0.91
R4a	9	96	13982.52	13725.92	1.87
R5a	11	120	16006.82	15736.66	1.72
R6a	13	144	20607.77	20465.39	0.70
R7a	4	36	5718.46	5610.05	1.93
R8a	6	72	11554.71	11343.19	1.86
R9a	8	108	17345.60	15632.09	10.96
R10a	10	144	23104.61	22430.00	3.01
R1b	3	24	3407.87	3379.74	0.83
R2b	5	48	5925.35	5889.56	0.61
R3b	7	72	11045.26	11006.12	0.36
R4b	9	96	12856.75	12807.87	0.38
R5b	11	120	14874.48	14544.13	2.27
R6b	13	144	18795.79	18518.82	1.50
R7b	4	36	5202.93	5136.37	1.30
R8b	6	72	10791.67	10703.17	0.83
R9b	8	108	15180.91	15013.71	1.11
R10b	10	144	20492.56	19969.15	2.62

Quality of service is provided through drastically reducing customer's inconvenience while maintaining low operational costs.

Table 2. Best results by simulated annealing.

Instance	Route Duration (min)	Waiting Time (min)		Ride Time (min)		CPU time (min)
		Total	Avg.	Total	Avg.	
R1a	831.3	98.51	2.05	241.93	10.08	1.00
R2a	1992.34	594.9	6.2	310.17	6.46	1.20
R3a	2404.67	132.93	0.92	894.08	12.42	1.46
R5a	3920.25	434.81	1.81	899.35	7.49	1.79
R9a	3258.66	34.42	0.16	1275.06	11.81	2.28
R10a	4475.42	203.33	0.71	2204.85	15.31	2.72
R1b	738.42	6.57	0.14	206.66	8.61	0.92
R2b	1428.44	31.75	0.33	311.95	6.5	1.30
R5b	3654.02	243.94	1.02	855.16	7.13	1.95
R6b	4318.33	149.02	0.52	1245.66	8.65	1.94
R7b	1095.67	0	0	345.1	9.59	1.05
R9b	3315.28	114.19	0.53	1085.18	10.05	2.26
R10b	4332.69	38.04	0.13	1427.08	9.91	2.77
TOTAL	35765.49	2082.41	14.52	11302.23	124.01	22.64

Table 3. Best results by genetic algorithm (Jorgensen et al., 2007).

Instance	Route Duration (min)	Waiting Time (min)		Ride Time (min)		CPU Time (min)
		Total	Avg.	Total	Avg.	
R1a	1039	260	5.42	310	12.90	5.57
R2a	1994	514	5.36	1330	27.72	11.43
R3a	2781	301	2.09	2894	40.20	21.58
R5a	4274	527	2.20	4837	40.30	58.23
R9a	3526	32	0.15	6719	62.21	40.78
R10a	5025	246	0.86	8341	57.92	65.98
R1b	928	164	3.42	549	22.89	5.46
R2b	1710	162	1.69	1300	27.07	11.72
R5b	4336	568	2.37	4720	39.33	58.93
R6b	5227	513	1.78	6397	44.42	81.23
R7b	1316	128	1.78	784	21.76	8.29
R9b	3676	177	0.82	5358	49.61	44.66
R10b	4678	85	0.29	8119	56.38	66.41
TOTAL	40508	3678	28.21	51657	502.72	488.61

The SA computational results produced good quality solutions for all the instances in expressively low computational times. These results were compared against a recent approach found in the literature, and in all of cases, the quality of service was better. The customers' inconvenience was significantly reduced presenting better solutions in practice. So, we can conclude based on the presented tests that SA maintain the low operational costs while drastically reducing the customers' inconvenience, improving the overall quality of service.

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